
Projective Geometry

Adapted from original slides in Computer Vision by Szeliski

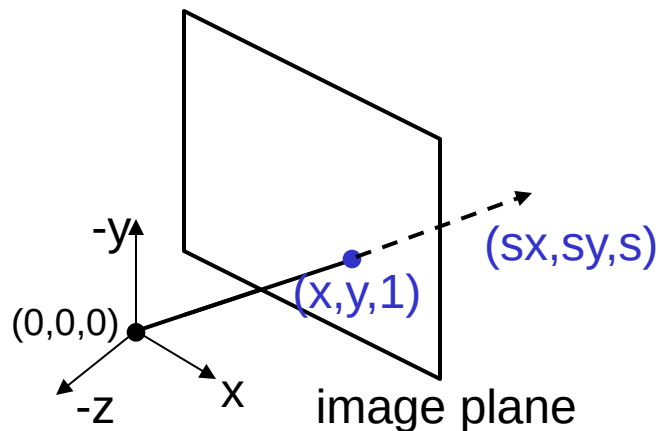
The projective plane

Why do we need homogeneous coordinates?

- represent points at infinity, homographies, perspective projection, multi-view relationships

What is the geometric intuition?

- a point in the image is a *ray* in projective space



- Each point (x,y) on the plane is represented by a ray (sx,sy,s)
 - all points on the ray are equivalent: $(x, y, 1) \cong (sx, sy, s)$

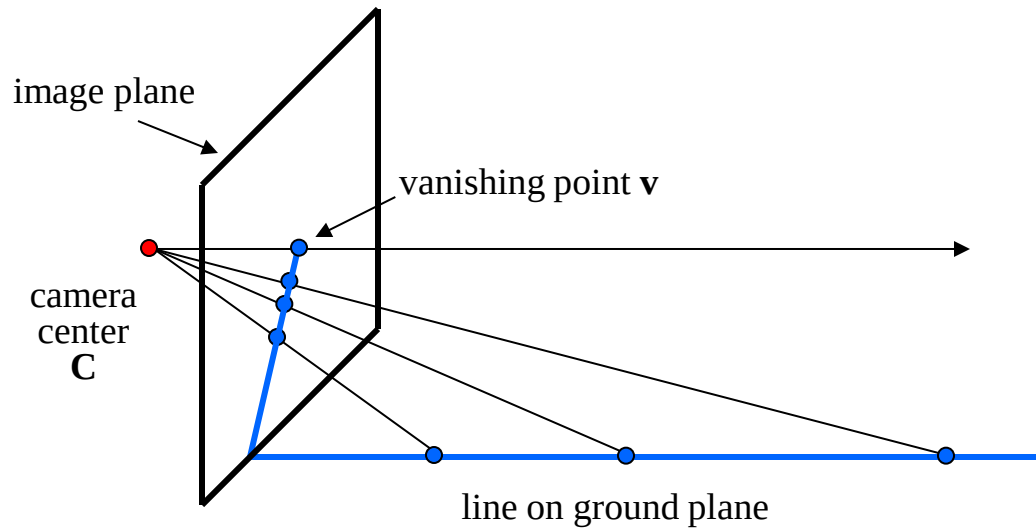
3D to 2D: “perspective” projection

Matrix Projection:
$$\mathbf{p} = \begin{bmatrix} wx \\ wy \\ w \end{bmatrix} = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \mathbf{P}\mathbf{P}$$

What is *not* preserved under perspective projection?

What IS preserved?

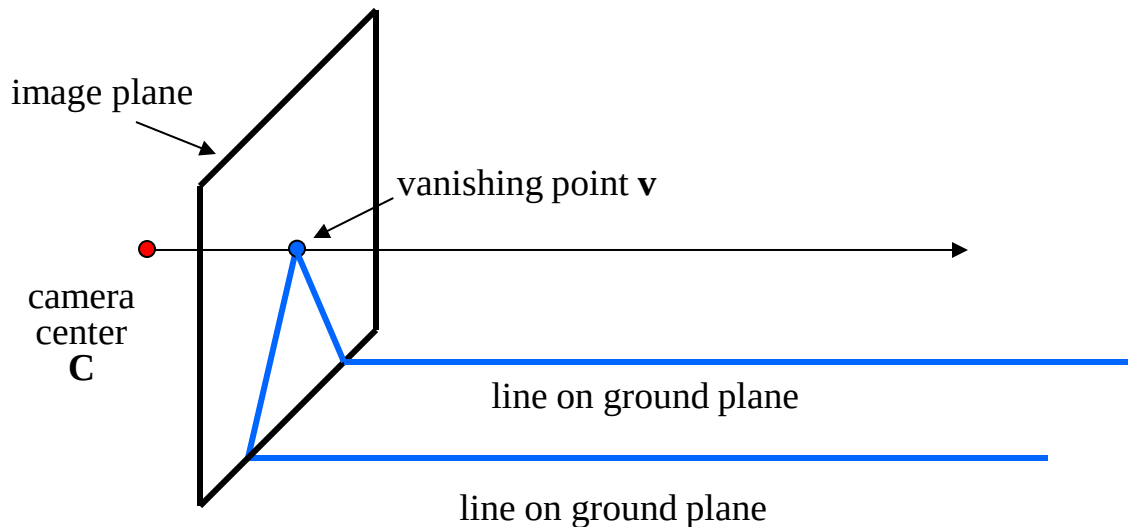
Vanishing points



Vanishing point

- projection of a point at infinity

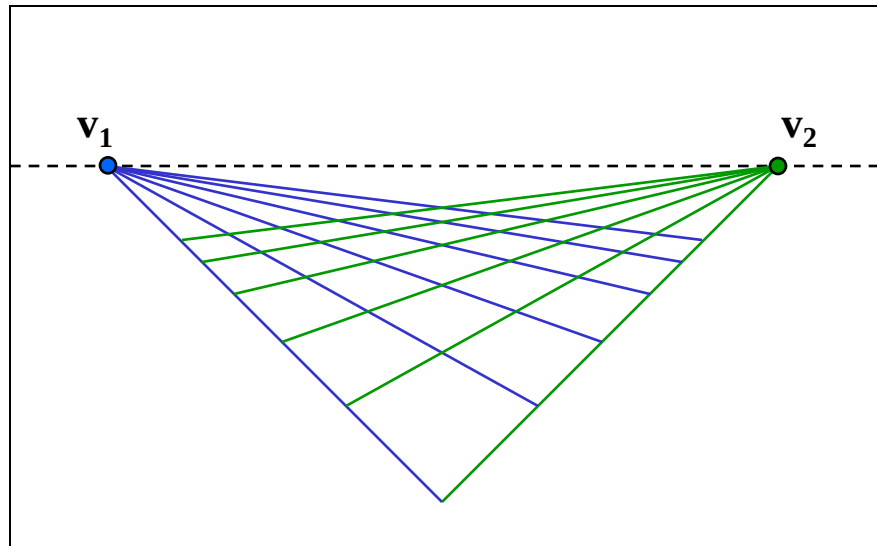
Vanishing points



Properties

- Any two parallel lines have the same vanishing point v
- The ray from C through v is parallel to the lines
- An image may have more than one vanishing point
 - in fact every pixel is a potential vanishing point

Vanishing lines



Multiple Vanishing Points

- Any set of parallel lines on the plane define a vanishing point
- The union of all of vanishing points from lines on the same plane is the *vanishing line*
 - For the ground plane, this is called the *horizon*

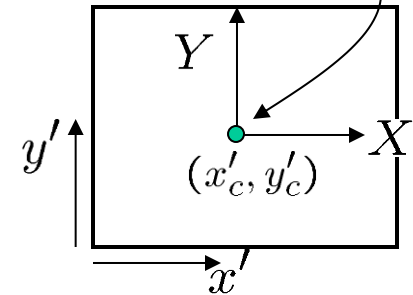
Camera parameters

A camera is described by several parameters

- Translation **T** of the optical center from the origin of world coords
- Rotation **R** of the image plane
- focal length **f**, principle point (x'_c, y'_c) , pixel size (s_x, s_y)
- blue parameters are called “**extrinsics**,” red are “**intrinsics**”

Projection equation

$$\mathbf{X} = \begin{bmatrix} sx \\ sy \\ s \end{bmatrix} = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \mathbf{\Pi} \mathbf{X}$$



- The projection matrix models the cumulative effect of all parameters
- Useful to decompose into a series of operations

$$\mathbf{\Pi} = \underbrace{\begin{bmatrix} -fs_x & 0 & x'_c \\ 0 & -fs_y & y'_c \\ 0 & 0 & 1 \end{bmatrix}}_{\text{intrinsics}} \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_{\text{projection}} \underbrace{\begin{bmatrix} \mathbf{R}_{3 \times 3} & \mathbf{0}_{3 \times 1} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix}}_{\text{rotation}} \underbrace{\begin{bmatrix} \mathbf{I}_{3 \times 3} & \mathbf{T}_{3 \times 1} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix}}_{\text{translation}}$$

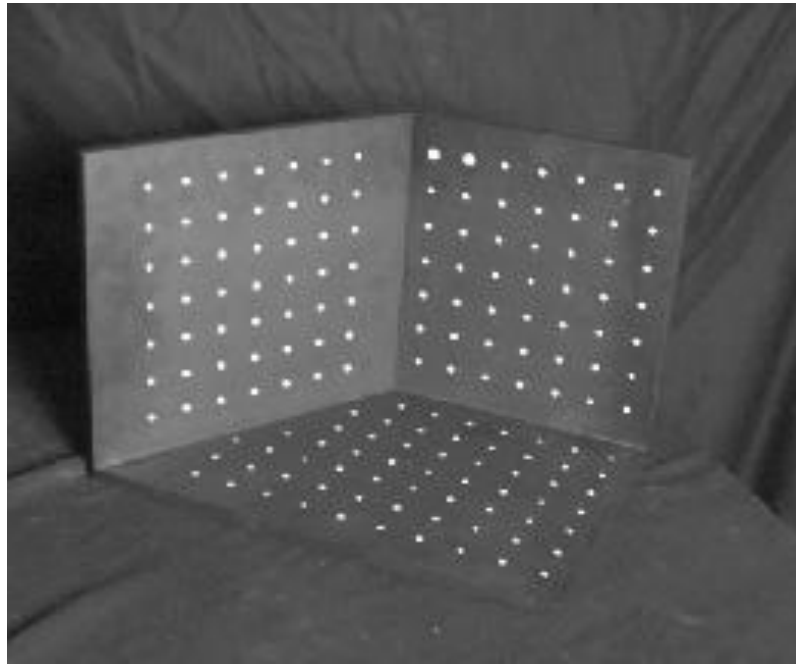
identity matrix

- The definitions of these parameters are **not** completely standardized
 - especially intrinsics—varies from one book to another

Estimating the projection matrix

Place a known object in the scene

- identify correspondence between image and scene
- compute mapping from scene to image



$$\begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix} \approx \begin{bmatrix} m_{00} & m_{01} & m_{02} & m_{03} \\ m_{10} & m_{11} & m_{12} & m_{13} \\ m_{20} & m_{21} & m_{22} & m_{23} \end{bmatrix} \begin{bmatrix} X_i \\ Y_i \\ Z_i \\ 1 \end{bmatrix}$$

Camera calibration

Goal: estimate the camera parameters

- Version 1: solve for projection matrix

$$\mathbf{X} = \begin{bmatrix} wx \\ wy \\ w \end{bmatrix} = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \mathbf{P}\mathbf{X}$$

- Version 2: solve for camera parameters separately
 - intrinsics (focal length, principle point, pixel size)
 - extrinsics (rotation angles, translation)
 - radial distortion

Direct linear calibration

$$\begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix} \cong \begin{bmatrix} m_{00} & m_{01} & m_{02} & m_{03} \\ m_{10} & m_{11} & m_{12} & m_{13} \\ m_{20} & m_{21} & m_{22} & m_{23} \end{bmatrix} \begin{bmatrix} X_i \\ Y_i \\ Z_i \\ 1 \end{bmatrix}$$

$$u_i = \frac{m_{00}X_i + m_{01}Y_i + m_{02}Z_i + m_{03}}{m_{20}X_i + m_{21}Y_i + m_{22}Z_i + m_{23}}$$

$$v_i = \frac{m_{10}X_i + m_{11}Y_i + m_{12}Z_i + m_{13}}{m_{20}X_i + m_{21}Y_i + m_{22}Z_i + m_{23}}$$

$$u_i(m_{20}X_i + m_{21}Y_i + m_{22}Z_i + m_{23}) = m_{00}X_i + m_{01}Y_i + m_{02}Z_i + m_{03}$$

$$v_i(m_{20}X_i + m_{21}Y_i + m_{22}Z_i + m_{23}) = m_{10}X_i + m_{11}Y_i + m_{12}Z_i + m_{13}$$

$$\begin{bmatrix} X_i & Y_i & Z_i & 1 & 0 & 0 & 0 & 0 & -u_iX_i & -u_iY_i & -u_iZ_i & -u_i \\ 0 & 0 & 0 & 0 & X_i & Y_i & Z_i & 1 & -v_iX_i & -v_iY_i & -v_iZ_i & -v_i \end{bmatrix} \begin{bmatrix} m_{00} \\ m_{01} \\ m_{02} \\ m_{03} \\ m_{10} \\ m_{11} \\ m_{12} \\ m_{13} \\ m_{20} \\ m_{21} \\ m_{22} \\ m_{23} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Direct linear calibration

$$\begin{bmatrix}
 X_1 & Y_1 & Z_1 & 1 & 0 & 0 & 0 & 0 & -u_1 X_1 & -u_1 Y_1 & -u_1 Z_1 & -u_1 \\
 0 & 0 & 0 & 0 & X_1 & Y_1 & Z_1 & 1 & -v_1 X_1 & -v_1 Y_1 & -v_1 Z_1 & -v_1 \\
 & & & & & & & \vdots & & & & \\
 X_n & Y_n & Z_n & 1 & 0 & 0 & 0 & 0 & -u_n X_n & -u_n Y_n & -u_n Z_n & -u_n \\
 0 & 0 & 0 & 0 & X_n & Y_n & Z_n & 1 & -v_n X_n & -v_n Y_n & -v_n Z_n & -v_n
 \end{bmatrix}
 \begin{bmatrix}
 m_{00} \\
 m_{01} \\
 m_{02} \\
 m_{03} \\
 m_{10} \\
 m_{11} \\
 m_{12} \\
 m_{13} \\
 m_{20} \\
 m_{21} \\
 m_{22}
 \end{bmatrix}
 =
 \begin{bmatrix}
 0 \\
 0 \\
 \vdots \\
 0 \\
 0
 \end{bmatrix}$$

Can solve for m_{ij} by linear least squares
 - look at solution for homographies.

Solving Homogeneous Linear Systems

The Core Concept: Constrained Optimization

If we want to minimize Ax but avoid the useless "trivial solution" ($x = 0$), we must constrain the size of x . The standard approach is to require x to be a unit vector, meaning its length is 1:

$$\|x\|^2 = x^T x = 1$$

To make Ax as close to the zero vector as possible, we minimize its squared Euclidean norm, $\|Ax\|^2$. Using linear algebra properties, we can rewrite this norm:

$$\|Ax\|^2 = (Ax)^T (Ax) = x^T A^T Ax$$

So, our goal is to find the vector x that minimizes $x^T A^T Ax$ subject to the constraint $x^T x = 1$.

Step 1: Compute the Matrix $A^T A$

Multiply the transpose of your coefficient matrix A by A itself.

- If A is an $m \times n$ matrix, $A^T A$ will always be a **square ($n \times n$), symmetric, and positive semi-definite** matrix.
- These properties are crucial because they guarantee that all eigenvalues of $A^T A$ are real numbers and are greater than or equal to zero.

Solving Homogeneous Linear Systems

Step 2: Set up the Eigenvalue Problem

To solve our constrained optimization problem, we use the method of **Lagrange Multipliers**.

We define the Lagrangian function $\mathcal{L}(x, \lambda)$ with a multiplier λ :

$$\mathcal{L}(x, \lambda) = x^T A^T A x - \lambda(x^T x - 1)$$

To find the minimum, we take the derivative with respect to x and set it to zero:

$$\frac{\partial \mathcal{L}}{\partial x} = 2A^T A x - 2\lambda x = 0$$

Dividing by 2 gives us the classic **eigenvalue equation**:

$$A^T A x = \lambda x$$

This proves that any valid solution x *must* be an eigenvector of $A^T A$.

Step 3: Identify the Minimum Eigenvalue

Now, let's substitute this back into our original minimization objective ($x^T A^T A x$):

$$x^T (A^T A x) = x^T (\lambda x) = \lambda(x^T x)$$

Since we constrained x to be a unit vector ($x^T x = 1$), the expression simplifies beautifully:

$$x^T A^T A x = \lambda$$

This means the value of our objective function is exactly equal to the eigenvalue λ . Therefore, to minimize the objective function, **we must choose the smallest possible eigenvalue, $\lambda_{\min}$** .

Solving Homogeneous Linear Systems

Step 4: Find the Corresponding Eigenvector

1. Find the smallest eigenvalue $\lambda_{\min}$ by solving the characteristic equation $\det(A^T A - \lambda I) = 0$ (or using numerical methods like the Power Method or SVD for large matrices).
2. Find the null space of $(A^T A - \lambda_{\min} I)$ to get the corresponding eigenvector x .
3. Normalize x so that $\|x\| = 1$. This vector x is your optimal solution.

Connection to Singular Value Decomposition (SVD)

In practice, computing $A^T A$ directly can introduce numerical instability.

The SVD of A is defined as:

$$A = U\Sigma V^T$$

Where:

- U contains the eigenvectors of AA^T .
- V contains the eigenvectors of $A^T A$.
- Σ contains the singular values (σ), which are the square roots of the eigenvalues of $A^T A$ ($\lambda = \sigma^2$).

Because the columns of V are sorted in order of descending singular values, **the very last column of V** corresponds to the smallest singular value (and thus the lowest eigenvalue of $A^T A$). This makes the last column of V the exact non-trivial solution to $Ax = 0$.

Direct linear calibration

Advantage:

- Very simple to formulate and solve

Disadvantages:

- Doesn't tell you the camera parameters
- Doesn't model radial distortion
- Hard to impose constraints (e.g., known focal length)
- Doesn't minimize the right error function

For these reasons, *nonlinear methods* are preferred

- Define error function E between projected 3D points and image positions
 - E is nonlinear function of intrinsics, extrinsics, radial distortion
- Minimize E using nonlinear optimization techniques
 - e.g., variants of Newton's method (e.g., Levenberg Marquart)

Camera calibration

Goal: estimate the camera parameters

- Version 1: solve for projection matrix

$$\mathbf{X} = \begin{bmatrix} wx \\ wy \\ w \end{bmatrix} = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \mathbf{P}\mathbf{X}$$

- Version 2: solve for camera parameters separately
 - intrinsics (focal length, principle point, pixel size)
 - extrinsics (rotation angles, translation)
 - radial distortion

(Slides 15-18: Self-study)

Finding the scale factors...

Let's assume that the camera is reasonable

- Square pixels
- Image plane parallel to sensor plane
- Principal point in the center of the image

$$\mathbf{\Pi} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1/f \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & t_1 \\ 0 & 1 & 0 & t_2 \\ 0 & 0 & 1 & t_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} r_{11} & r_{12} & r_{13} & t'_1 \\ r_{21} & r_{22} & r_{23} & t'_2 \\ r_{31}/f & r_{32}/f & r_{33}/f & t'_3/f \end{bmatrix} \quad \begin{bmatrix} t'_1 \\ t'_2 \\ t'_3 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} t_1 \\ t_2 \\ t_3 \end{bmatrix}$$

$$= \begin{bmatrix} a \mathbf{v}_X & b \mathbf{v}_Y & c \mathbf{v}_Z & d \mathbf{o} \end{bmatrix}$$

Solving for f

$$\begin{bmatrix} v_{x1} & v_{y1} & v_{z1} & o_1 \\ v_{x2} & v_{y2} & v_{z2} & o_2 \\ v_{x3} & v_{y3} & v_{z3} & o_3 \end{bmatrix} = \begin{bmatrix} r_{11}/a & r_{12}/b & r_{13}/c & t'_1/d \\ r_{21}/a & r_{22}/b & r_{23}/c & t'_2/d \\ r_{31}/af & r_{32}/bf & r_{33}/cf & t'_3/df \end{bmatrix}$$

$$\begin{bmatrix} v_X & v_Y & v_Z & o \end{bmatrix}$$

$$\begin{bmatrix} v_{x1} & v_{y1} & v_{z1} & o_1 \\ v_{x2} & v_{y2} & v_{z2} & o_2 \\ fv_{x3} & fv_{y3} & fv_{z3} & fo_3 \end{bmatrix} = \begin{bmatrix} r_{11}/a & r_{12}/b & r_{13}/c & t'_1/d \\ r_{21}/a & r_{22}/b & r_{23}/c & t'_2/d \\ r_{31}/a & r_{32}/b & r_{33}/c & t'_3/d \end{bmatrix}$$

Orthogonal
vectors

Orthogonal
vectors

$$v_{x1}v_{y1} + v_{x2}v_{y2} + f^2v_{x3}v_{y3} = 0 \quad \longrightarrow \quad f = \sqrt{\frac{v_{x1}v_{y1} + v_{x2}v_{y2}}{-v_{x3}v_{y3}}}$$

Solving for a, b, and c

$$\begin{bmatrix} v_{x1}/f & v_{y1}/f & v_{z1}/f & o_1/f \\ v_{x2}/f & v_{y2}/f & v_{z2}/f & o_2/f \\ v_{x3} & v_{y3} & v_{z3} & o_3 \end{bmatrix} = \begin{bmatrix} r_{11}/a & r_{12}/b & r_{13}/c & t'_1/d \\ r_{21}/a & r_{22}/b & r_{23}/c & t'_2/d \\ r_{31}/a & r_{32}/b & r_{33}/c & t'_3/d \end{bmatrix}$$

Norm
= 1/a

Norm
= 1/a

Solve for a, b, c

- Divide the first two rows by f, now that it is known
- Now just find the norms of the first three columns
- Once we know a, b, and c, that also determines R

How about d?

- Need a reference point in the scene

Solving for d

$$\begin{bmatrix} wu \\ wv \\ w \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t'_1/d \\ r_{21} & r_{22} & r_{23} & t'_2/d \\ r_{31} & r_{32} & r_{33} & t'_3/d \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ H \\ 1 \end{bmatrix}$$

Suppose we have one reference height H

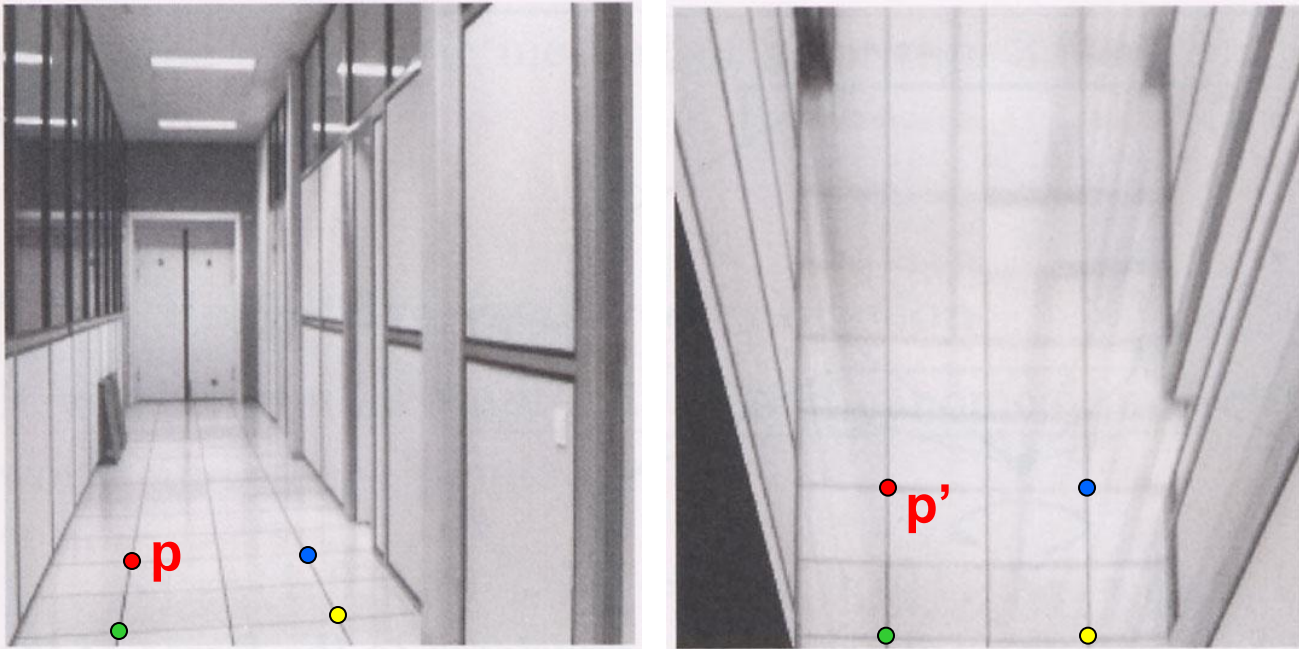
- E.g., we know that (0, 0, H) gets mapped to (u, v)

$$u = \frac{r_{13}H + t'_1/d}{r_{33}H + t'_3/d} \qquad d = \frac{t'_1 - ut'_3}{ur_{33}H - r_{13}H}$$

Finally, we can solve for t

$$\begin{bmatrix} t_1 \\ t_2 \\ t_3 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}^T \begin{bmatrix} t'_1 \\ t'_2 \\ t'_3 \end{bmatrix}$$

Image rectification



To unwarp (rectify) an image

- solve for homography $\mathbf{H}$ given $\mathbf{p}$ and $\mathbf{p}'$
- solve equations of the form: $w\mathbf{p}' = \mathbf{H}\mathbf{p}$
 - linear in unknowns: w and coefficients of $\mathbf{H}$
 - $\mathbf{H}$ is defined up to an arbitrary scale factor
 - how many points are necessary to solve for $\mathbf{H}$?

Solving for homographies

$$\begin{bmatrix} x'_i \\ y'_i \\ 1 \end{bmatrix} \cong \begin{bmatrix} h_{00} & h_{01} & h_{02} \\ h_{10} & h_{11} & h_{12} \\ h_{20} & h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix}$$

$$x'_i = \frac{h_{00}x_i + h_{01}y_i + h_{02}}{h_{20}x_i + h_{21}y_i + h_{22}}$$

$$y'_i = \frac{h_{10}x_i + h_{11}y_i + h_{12}}{h_{20}x_i + h_{21}y_i + h_{22}}$$

$$x'_i(h_{20}x_i + h_{21}y_i + h_{22}) = h_{00}x_i + h_{01}y_i + h_{02}$$

$$y'_i(h_{20}x_i + h_{21}y_i + h_{22}) = h_{10}x_i + h_{11}y_i + h_{12}$$

$$\begin{bmatrix} x_i & y_i & 1 & 0 & 0 & 0 & -x'_i x_i & -x'_i y_i & -x'_i \\ 0 & 0 & 0 & x_i & y_i & 1 & -y'_i x_i & -y'_i y_i & -y'_i \end{bmatrix} \begin{bmatrix} h_{00} \\ h_{01} \\ h_{02} \\ h_{10} \\ h_{11} \\ h_{12} \\ h_{20} \\ h_{21} \\ h_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Solving for homographies

$$\begin{array}{c}
 \begin{bmatrix}
 x_1 & y_1 & 1 & 0 & 0 & 0 & -x'_1 x_1 & -x'_1 y_1 & -x'_1 \\
 0 & 0 & 0 & x_1 & y_1 & 1 & -y'_1 x_1 & -y'_1 y_1 & -y'_1 \\
 & & & & & \vdots & & & \\
 x_n & y_n & 1 & 0 & 0 & 0 & -x'_n x_n & -x'_n y_n & -x'_n \\
 0 & 0 & 0 & x_n & y_n & 1 & -y'_n x_n & -y'_n y_n & -y'_n
 \end{bmatrix}
 \begin{bmatrix}
 h_{00} \\
 h_{01} \\
 h_{02} \\
 h_{10} \\
 h_{11} \\
 h_{12} \\
 h_{20} \\
 h_{21} \\
 h_{22}
 \end{bmatrix}
 =
 \begin{bmatrix}
 0 \\
 0 \\
 \vdots \\
 0 \\
 0
 \end{bmatrix}
 \\
 \mathbf{A} \qquad \qquad \mathbf{h} \qquad \qquad \mathbf{0} \\
 2n \times 9 \qquad \qquad 9 \qquad \qquad 2n
 \end{array}$$

Defines a least squares problem: minimize $\|\mathbf{A}\mathbf{h} - \mathbf{0}\|^2$

- Since $\mathbf{h}$ is only defined up to scale, solve for unit vector $\hat{\mathbf{h}}$
- Solution: $\hat{\mathbf{h}}$ = eigenvector of $\mathbf{A}^\top \mathbf{A}$ with smallest eigenvalue
- Works with 4 or more points